



VP AIR TEST NEET (2025-26)

PART TEST - 02

DROPPER

DURATION: 180 Minutes

Date: 07-09-2025

M. MARKS: 720

ANSWER KEY

1. (1)	46. (4)	91. (2)	136. (3)
2. (2)	47. (1)	92. (3)	137. (3)
3. (3)	48. (3)	93. (3)	138. (3)
4. (1)	49. (3)	94. (3)	139. (1)
5. (4)	50. (1)	95. (1)	140. (1)
6. (3)	51. (4)	96. (2)	141. (2)
7. (3)	52. (1)	97. (3)	142. (3)
8. (2)	53. (1)	98. (4)	143. (2)
9. (4)	54. (2)	99. (4)	144. (3)
10. (3)	55. (4)	100. (1)	145. (2)
11. (2)	56. (3)	101. (3)	146. (4)
12. (2)	57. (3)	102. (1)	147. (3)
13. (2)	58. (2)	103. (3)	148. (2)
14. (3)	59. (4)	104. (2)	149. (4)
15. (1)	60. (3)	105. (2)	150. (3)
16. (4)	61. (3)	106. (2)	151. (3)
17. (4)	62. (3)	107. (1)	152. (1)
18. (2)	63. (4)	108. (1)	153. (3)
19. (3)	64. (2)	109. (3)	154. (4)
20. (4)	65. (1)	110. (2)	155. (1)
21. (2)	66. (3)	111. (3)	156. (3)
22. (2)	67. (2)	112. (1)	157. (3)
23. (1)	68. (1)	113. (3)	158. (3)
24. (3)	69. (4)	114. (4)	159. (2)
25. (4)	70. (1)	115. (3)	160. (3)
26. (2)	71. (3)	116. (3)	161. (1)
27. (2)	72. (3)	117. (3)	162. (3)
28. (2)	73. (1)	118. (2)	163. (3)
29. (1)	74. (4)	119. (3)	164. (3)
30. (2)	75. (3)	120. (2)	165. (3)
31. (4)	76. (4)	121. (1)	166. (1)
32. (1)	77. (4)	122. (2)	167. (3)
33. (3)	78. (2)	123. (2)	168. (3)
34. (1)	79. (3)	124. (1)	169. (1)
35. (3)	80. (3)	125. (3)	170. (1)
36. (2)	81. (3)	126. (3)	171. (3)
37. (2)	82. (1)	127. (4)	172. (4)
38. (4)	83. (4)	128. (2)	173. (4)
39. (1)	84. (4)	129. (1)	174. (3)
40. (3)	85. (2)	130. (4)	175. (3)
41. (2)	86. (3)	131. (1)	176. (3)
42. (1)	87. (3)	132. (3)	177. (2)
43. (2)	88. (4)	133. (4)	178. (2)
44. (1)	89. (1)	134. (1)	179. (3)
45. (4)	90. (3)	135. (2)	180. (3)

HINTS & SOLUTIONS**1. (1)**

$$v = u + at$$

$$\Rightarrow 40 = 0 + a \times 2 \Rightarrow a = 20 \text{ m/s}^2$$

$$v^2 = u^2 + 2ah \Rightarrow (40)^2 = 0 + 2 \times 20h$$

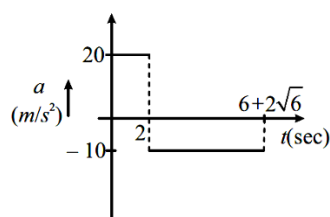
$$\Rightarrow h = \frac{40 \times 40}{40} = 40 \text{ m}$$

Time taken in reaching from $h = 40 \text{ m}$ to ground

$$h = ut - \frac{1}{2}gt^2 \Rightarrow -40 = 40t - \frac{1}{2} \times 10 \times t^2$$

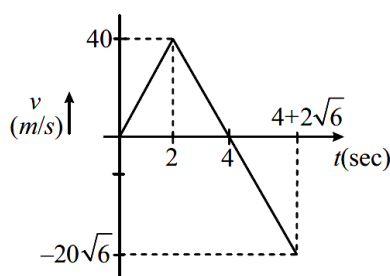
On solving

$$t = 4 + 2\sqrt{6} \text{ sec}$$



$$v^2 = 40^2 + 2(-10)(-40)$$

$$\Rightarrow v = -20\sqrt{6} \text{ m/s}$$



Therefore, option (1) is the correct answer.

(NCERT Ed. 2025-26, 11th, Page No. 19)**2. (2)**For the particles to collide, their vertical positions must be equal at the time of collision ($t = 0.5 \text{ s}$).Vertical position of particle A at time t :

$$y_A = (20\sqrt{3} \times \sin(\theta)) \times t - (1/2)gt^2$$

Vertical position of particle B at time t :

$$y_B = (20) \times t - (1/2)gt^2$$

At the time of collision ($t = 0.5 \text{ s}$), we must have

$$y_A = y_B:$$

$$(20\sqrt{3} \times \sin(\theta)) \times 0.5 - (1/2)g(0.5)^2$$

$$= (20) \times 0.5 - (1/2)g(0.5)^2$$

$$\sin(\theta) = 1/\sqrt{3}$$

Next, their horizontal positions must also be equal at the time of collision.

Horizontal position of particle A at time t :

$$x_A = (v_A \times \cos(\theta)) \times t$$

$$\text{At } t = 0.5 \text{ s, } x_A = (20\sqrt{3} \times \cos(\theta)) \times 0.5$$

$$x_A = 10\sqrt{3} \cos(\theta)$$

Particle B is projected vertically, so its horizontal velocity is zero. Its horizontal position remains constant.

$$x_B = S$$

For a collision to occur, $x_A = x_B = S$.

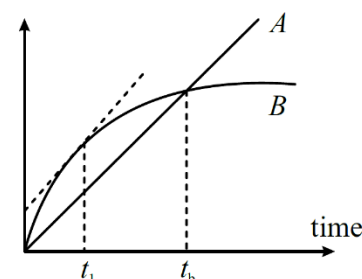
$$S = 10\sqrt{3} \cos(\theta)$$

$$\cos^2(\theta) = 1 - (1/\sqrt{3})^2 = 1 - 1/3 = 2/3$$

$$\cos(\theta) = \sqrt{(2/3)}$$

$$S = 10\sqrt{3} \times \sqrt{(2/3)}$$

$$S = 10\sqrt{2} \text{ meters}$$

(NCERT Ed. 2025-26, 11th, Page No. 39)**3. (3)**Both have same velocity at some time before t_b at time t_1 where both graphs have same slope.**(NCERT Ed. 2025-26, 11th, Page No. 14)****4. (1)**

$$(\vec{A} + \vec{B} + \vec{C}) = (3\hat{i} - 5\hat{k})$$

$$(\vec{A} \times \vec{B}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ -1 & 1 & 4 \end{vmatrix} = 5\hat{i} - 7\hat{j} + 3\hat{k}$$

$$\cos \theta = \frac{15 - 15}{|3\hat{i} - 5\hat{k}| |5\hat{i} - 7\hat{j} + 3\hat{k}|}$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

(NCERT Ed. 2025-26, 11th, Page No. 33)

5. (4)

$a = \frac{d^2x}{dt^2}$ is constant for particle 3 and 4.

(NCERT Ed. 2025-26, 11th, Page No. 16)

6. (3)

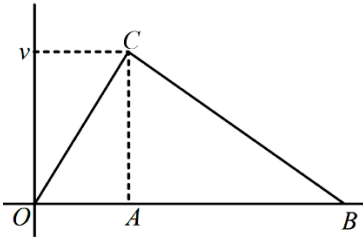
$$\text{Avg. velocity for } OA = \frac{\frac{1}{2} \times OA \times AC}{OA} = \frac{AC}{2}$$

$$\text{For } OB \quad V_{avg} = \frac{\frac{1}{2} \times AB \times AC}{AB} = \frac{AC}{2}$$

$$\frac{AC}{OA} = \sqrt{3} \Rightarrow \frac{AC}{AB} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{OA}{AB} = \frac{1}{3} \Rightarrow \frac{OA}{OB} = \frac{1}{4}$$

$$\frac{\text{Area } OAC}{\text{Area } OCB} = \frac{\frac{1}{2} \times AC \times OA}{\frac{1}{2} \times AC \times OB} = \frac{1}{4}$$

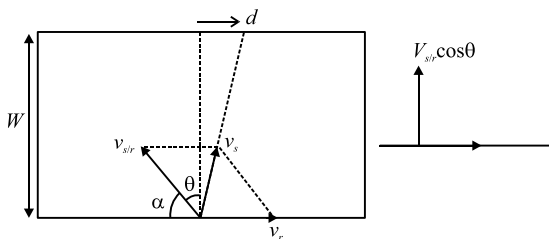


$$(a_{OA})_{avg} = \frac{AC}{OA}$$

$$(a_{OB})_{avg} = 0$$

(NCERT Ed. 2025-26, 11th, Page No. 17)

7. (3)



$\vec{V}_r - \vec{V}_{s/r} \rightarrow$ Causes drift

$$d = (V_r - V_{s/r} \sin \theta) \cdot t$$

If $V_r > V_{s/r}$, d can never be zero:

$$\text{So, } d = (V_r - V_{s/r} \sin \theta) \cdot \frac{W}{V_{s/r} \cos \theta}$$

$\therefore$ condⁿ for $d_{\min}$?

Differentiate w.r.t. and equate to zero.

$$\text{We get: } \sin \theta = \frac{V_{s/r}}{V_r} \Rightarrow \theta = \sin^{-1} \left(\frac{V_{s/r}}{V_r} \right)$$

$$\theta = \sin^{-1} \left(\frac{3}{5} \right)$$

$$\theta = 37^\circ$$

$$\alpha = 90^\circ - 37^\circ = 53^\circ$$

(NCERT Ed. 2025-26, 11th, Page No. 31)

8. (2)

(A) AB = maximum height

$$\frac{v^2}{2a} = \frac{20 \times 20}{2 \times 10} = 20 \text{ m}$$

$$\therefore \frac{BD}{AD} = \frac{20 + 20}{20} = 2$$

$$(B) \frac{s(1^{\text{th}})}{s(2^{\text{nd}})} = \frac{20 \times 1 - \frac{1}{2} \times 10 \times 1^2}{20 - [15]} = \frac{15}{5} = 3$$

$$(C) \frac{V_A}{V_B} = \frac{20}{\sqrt{20^2 + 2 \times 10 \times 20}} = \frac{20}{\sqrt{20^2 + 400}}$$

$$= \frac{20}{\sqrt{400 + 400}} = \frac{20}{\sqrt{800}} = \frac{20}{20\sqrt{2}} = \frac{1}{\sqrt{2}}$$

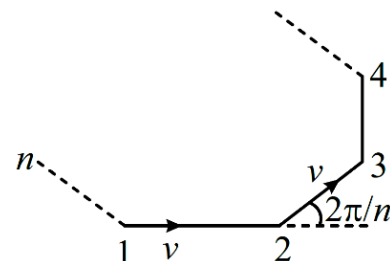
$$(D) \frac{t_{AC}}{t_{AB}} = \frac{2t_{AB}}{t_{AB}} = 2$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

9. (4)

$$t = \frac{a}{v - v \cos \frac{2\pi}{n}}$$

Here $n = 12$



$$\Rightarrow t = \frac{a}{v - v \cos \frac{2\pi}{12}} = \frac{2a}{v(2 - \sqrt{3})}$$

(NCERT Ed. 2025-26, 11th, Page No. 32)

10. (3)

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

$$b = \frac{\text{total area}}{\text{total time}} = \frac{10+20}{6} = \frac{30}{6} = 5 \text{ m/s}$$

Average acceleration

$$= \frac{\text{change in velocity}}{\text{time taken}}$$

$$C = \frac{10 - (-10)}{4} = \frac{20}{4} = 5 \text{ m/s}^2$$

$$bc = (5)(5) = 25 \text{ m}^2/\text{s}^3$$

(NCERT Ed. 2025-26, 11th, Page No. 25)

11. (2)

$$v \frac{dv}{dx} = 4 - 2x$$

$$\int_0^v v dv = \int_0^x (4 - 2x) dx \Rightarrow \frac{v^2}{2} = 4x - x^2$$

$$\text{When } v = 0, 4x - x^2 = 0$$

$$x = 0, 4$$

$\therefore$ At $x = 4 \text{ m}$, the particle will again come to rest.

(NCERT Ed. 2025-26, 11th, Page No. 18)

12. (2)

$$x = \frac{1}{2} at^2 \quad \dots(i)$$

$$y = ut + \frac{1}{2} at^2 \quad \dots(ii)$$

From (i) and (ii)

$$y = u \sqrt{\frac{2x}{a}} + x \Rightarrow (y - x)^2 = \frac{2u^2 x}{a}$$

(NCERT Ed. 2025-26, 11th, Page No. 38)

13. (2)

$$V = V_0 + gt + Ft^2$$

$$\frac{dx}{dt} = V_0 + gt + Ft^2$$

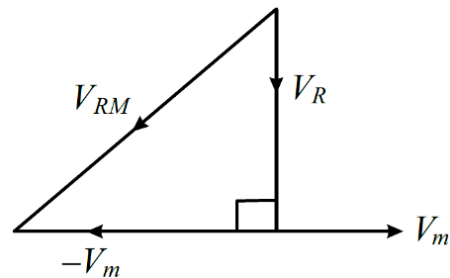
$$\Rightarrow \int_{x=0}^x dx = \int_{t=0}^{t=1} (V_0 + gt + Ft^2) dt$$

$$x = \left[V_0 t + \frac{gt^2}{2} + \frac{Ft^3}{3} \right]_{t=0}^{t=1}$$

$$\Rightarrow x = V_0 + \frac{g}{2} + \frac{F}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

14. (3)



$$\vec{V}_{RM} = \vec{V}_R - \vec{V}_m$$

$$V_{RM} = \sqrt{V_R^2 + V_m^2} = \sqrt{109} \text{ km/h}$$

(NCERT Ed. 2025-26, 11th, Page No. 30)

15. (1)

$$\vec{v}_A = 400\hat{i} \quad \dots(i)$$

$$\vec{v}_B = v_B \cos 30^\circ \hat{i} + v_B \sin 30^\circ \hat{j}$$

$$\vec{v}_B = \frac{v_B \sqrt{3}}{2} \hat{i} + \frac{v_B}{2} \hat{j} \quad \dots(ii)$$

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A$$

$$\vec{v}_{B/A} = \left(\frac{v_B \sqrt{3}}{2} \hat{i} + \frac{v_B}{2} \hat{j} \right) - 400\hat{i}$$

$$\vec{v}_{B/A} = \left(\frac{v_B \sqrt{3}}{2} - 400 \right) \hat{i} + \frac{v_B}{2} \hat{j}$$

x component of $v_{B/A} = 0$

$$\frac{v_B \sqrt{3}}{2} - 400 = 0 \Rightarrow v_B = \frac{800}{\sqrt{3}}$$

$$\vec{v}_B = 400\hat{i} + \frac{400}{\sqrt{3}} \hat{j}$$

(NCERT Ed. 2025-26, 11th, Page No. 33)

16. (4)

Change in magnitude of velocity = 0

(NCERT Ed. 2025-26, 11th, Page No. 40)

17. (4)

Maximum acceleration means maximum change in velocity in minimum time interval.

In time interval $t = 30$ to $t = 40$ sec

$$a = \frac{\Delta v}{\Delta t} = \frac{80 - 20}{40 - 30} = \frac{60}{10} = 6 \text{ cm/sec}^2.$$

(NCERT Ed. 2025-26, 11th, Page No. 22)

18. (2)

$$1^{\text{st}} \text{ stone } h = -ut_1 + \frac{1}{2}gt_1^2 \quad \dots(i)$$

$$2^{\text{nd}} \text{ stone } h = ut_2 + \frac{1}{2}gt_2^2 \quad \dots(ii)$$

Equation (i) $\times t_2$ and equation (ii) $\times t_1$ and added

$$h(t_1 + t_2) = g(t_1 t_2)(t_1 + t_2)$$

$$\Rightarrow h = \frac{1}{2}g(t_1 t_2)$$

$$\text{For } 3^{\text{rd}} \text{ stone } t = \sqrt{t_1 t_2}$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

19. (3)

$$\frac{dv}{dt} = 6 - 3v \Rightarrow \frac{dv}{6 - 3v} = dt$$

Integrating both sides,

$$\int \frac{dv}{6 - 3v} = \int dt$$

$$\Rightarrow \frac{\log_e(6 - 3v)}{-3} = t + K_1$$

$$\Rightarrow \log_e(6 - 3v) = -3t + K_2 \quad \dots(i)$$

$$\text{At } t = 0, v = 0 \Rightarrow \log_e 6 = K_2$$

Substituting the value of K_2 in equation (i)

$$\log_e(6 - 3v) = -3t + \log_e 6$$

$$\Rightarrow \log_e \left(\frac{6 - 3v}{6} \right) = -3t$$

$$\Rightarrow e^{-3t} = \frac{6 - 3v}{6}$$

$$\Rightarrow 6 - 3v = 6e^{-3t} \Rightarrow 3v = 6(1 - e^{-3t})$$

$$\Rightarrow v = 2(1 - e^{-3t})$$

$$\therefore v_{\text{terminal}} = 2 \text{ m/s (When } t = \infty)$$

Acceleration

$$a = \frac{dv}{dt} = \frac{d}{dt} [2(1 - e^{-3t})] = 6e^{-3t}$$

$$\text{Initial acceleration} = 6 \text{ m/s}^2.$$

$$\text{So, } 3 = 6e^{-3t}$$

$$\Rightarrow e^{-3t} = \frac{1}{2}$$

$$\text{Thus, } v = 2 \left(1 - \frac{1}{2} \right)$$

$$= 1 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 16)

20. (4)

The vector $\hat{i} + x\hat{j} + 3\hat{k}$ is doubled in magnitude.

$$\therefore 2|\hat{i} + x\hat{j} + 3\hat{k}|$$

$$= |4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}|$$

$$\Rightarrow 2\sqrt{1^2 + x^2 + 3^2}$$

$$= \sqrt{4^2 + (4x - 2)^2 + 2^2}$$

$$\Rightarrow 4(10 + x^2) = 24 + 16x^2 - 16x + 4$$

$$\Rightarrow 40 + 4x^2 = 20 + 16x^2 - 16x + 4$$

$$\Rightarrow 12x^2 - 16x - 16 = 0 \Rightarrow (x - 2)$$

$$(3x^2 + 2) = 0$$

$$\therefore x = 2, x = -\frac{2}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 33)

21. (2)

As $\theta \uparrow$, H and T both increase.

But $R \uparrow$ from 0° to 45° , at $\theta = 45^\circ$ becomes maximum, then decreases.

So, between 30° to $45^\circ R \uparrow$, then between 45° to $60^\circ R \downarrow$.

(NCERT Ed. 2025-26, 11th, Page No. 40)

22. (2)

$$S_n = u + \frac{a}{2}(2n - 1),$$

$$\Rightarrow 102.5 = u + \frac{a}{2}(2n - 1) \text{ \&}$$

$$115 = u + \frac{a}{2}(2n + 3)$$

$$\Rightarrow 12.5 = 2a \Rightarrow a = 6.25 \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

23. (1)

Due north will take him to cross in shortest time.

(NCERT Ed. 2025-26, 11th, Page No. 31)

24. (3)

Angular velocity of every particle of disc is same

$$a_p = \omega^2 r_p, a_Q = \omega^2 r_Q$$

$$\therefore r_p > r_Q \Rightarrow a_p > a_Q$$

(NCERT Ed. 2025-26, 11th, Page No. 42)

25. (4)

$$X = R + \frac{1}{2} \left(\frac{g}{2} \right) \left[\frac{2u \sin \theta}{g} \right]^2$$

$$= R + \frac{g}{4} \left[\frac{4u^2 \sin^2 \theta}{g} \right]$$

$$= R + \frac{u^2 \sin^2 \theta}{g} = R + 2H$$

(NCERT Ed. 2025-26, 11th, Page No. 38)

26. (2)

$$\sin 30^\circ = \frac{v_y}{v}$$

$$\frac{1}{2} = \frac{(20 \sin 60^\circ)}{v} \Rightarrow v = 20\sqrt{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 38)

27. (2)

$$a_y = -3 \text{ m/s}^2 \text{ and } a_x = 2 \text{ m/s}^2$$

It has maximum y -coordinate, when

$$v_y = 0 = 6 - 3(t)$$

$$\therefore t = 2 \text{ sec,}$$

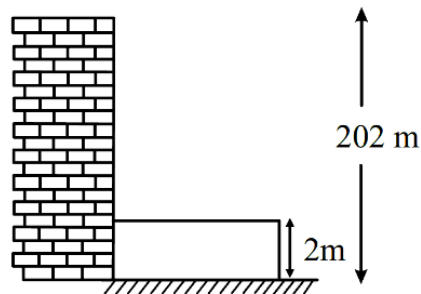
at that instant,

$$y = 6 \times 2 - \frac{1}{2} \times 3 \times 2^2 = 6 \text{ m}$$

$$\text{and } x = \frac{1}{2} \times 2 \times 2^2 = 4 \text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 38)

28. (2)



$$v_0^2 - v^2 = 2ah = \sqrt{2ax} = \sqrt{2(10)200}$$

$$v^2 - v_0^2 = 2a(x - x_0)$$

$$0^2 - 4000 = 2a(0.5 - 2)$$

$$a = \frac{4000}{3} \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

29. (1)

$$t_0 = \sqrt{\frac{2h}{g}}$$

$$\Rightarrow V_0 = \sqrt{\frac{2h}{g}} \times g = \sqrt{2gh}$$

$$\therefore t_1 = \text{time to reach top} = \frac{v_0}{g}$$

$$\Rightarrow H = h + h' = 2h$$

$$\therefore t_2 = \text{time of fall}$$

$$= \sqrt{\frac{2 \times (2h)}{g}} = 2\sqrt{\frac{h}{g}}$$

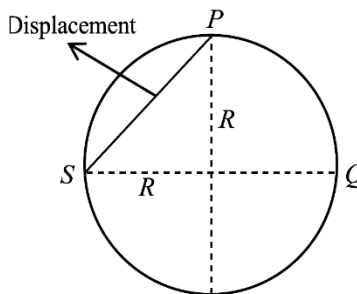
$$\therefore \text{Total time} = t_1 + t_2$$

$$= (2 + \sqrt{2})\sqrt{\frac{h}{g}} = 3.4\sqrt{\frac{h}{g}}$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

30. (2)

Given: $R = 2 \text{ km}$



$$\therefore \text{Displacement} = PS$$

$$\sqrt{R^2 + R^2} = R\sqrt{2} = 2\sqrt{2} = \sqrt{8} \text{ km}$$

(NCERT Ed. 2025-26, 11th, Page No. 28)

31. (4)

At $t = 0$, $v = 0 \text{ m/s}$

From 0 to $t_1 \rightarrow$ acceleration is constant and positive, hence slope of v - t graph should be constant and positive.

From t_1 to $t_2 \rightarrow$ acceleration is zero, hence slope of v - t graph should be zero.

(NCERT Ed. 2025-26, 11th, Page No. 16)

32. (1)

To avoid an accident, bike B must overtake bike A before bike A and bike C meet.

Relative speed of A and C

$$= v_A + v_C = 15 \text{ m/s} + 20 \text{ m/s} = 35 \text{ m/s}.$$

Time (t) = $1000 \text{ m} / 35 \text{ m/s} = 200/7$ seconds.

Bike B must cover the 200 m distance to bike A within this time.

To analyze the overtake, we consider the motion of B relative to A .

Initial relative speed (u_{rel})

$$= v_B - v_A = 15 \text{ m/s} - 15 \text{ m/s} = 0 \text{ m/s}.$$

Relative distance (s_{rel}) = 200 m .

$$200 = (0 \times t) + (1/2) \times a \times (200/7)^2$$

$$200 = (1/2) \times a \times (40000/49)$$

$$400 = a \times (40000/49)$$

$$a = (400 \times 49) / 40000$$

$$a = 49/100 = 0.49 \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

33. (3)

$\therefore$ Acceleration down the plane = $g \sin \theta$

$$\text{Since } l = 0 + \frac{1}{2} g \sin \theta t^2$$

$$\therefore t^2 = \frac{2l}{g \sin \theta} = \frac{2h}{g \sin^2 \theta}$$

$$\Rightarrow t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

34. (1)

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 2}{10}} = \frac{2}{\sqrt{10}} \text{ sec.}$$

$$v = \frac{R}{t}$$

$$\Rightarrow 5.1 \times \frac{\sqrt{10}}{2} = 8.1 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 39)

35. (3)

Let P be the smaller force and Q be the greater force then according to problem: $P + Q = 18 \dots(i)$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta} = 12 \dots(ii)$$

$$\tan \phi = \frac{Q \sin \theta}{P + Q \cos \theta} = \tan 90^\circ = \infty$$

$$\therefore P + Q \cos \theta = 0 \dots(iii)$$

By solving (i), (ii) and (iii)

we will get $P = 5$ and $Q = 13$.

Therefore, option (3) is the correct answer.

(NCERT Ed. 2025-26, 11th, Page No. 34)

36. (2)

Find Rain's Horizontal Speed (V_x):

At $t = 4 \text{ s}$, the rain appears vertical. This means the man's speed exactly cancels out the rain's horizontal speed.

Man's speed at 4s:

$$V_m = a \times t = 1.5 \text{ m/s}^2 \times 4 \text{ s} = 6 \text{ m/s}.$$

Therefore, the rain's horizontal speed is $V_x = 6 \text{ m/s}$

Set Condition for 45° Angle:

The rain appears at 45° when the magnitude of the relative horizontal speed equals the rain's vertical speed.

Initially at 45° , we know $V_x = V_y$, so the rain's vertical speed $V_y = 6 \text{ m/s}$.

At a new time t' , the man's speed is $V_{m'} = 1.5t'$.

The condition is: $|V_x - V_{m'}| = V_y \Rightarrow |6 - 1.5t'| = 6$.

Solve for Time (t'):

For the non-trivial solution (where $t' > 0$), the expression inside the absolute value must be negative:

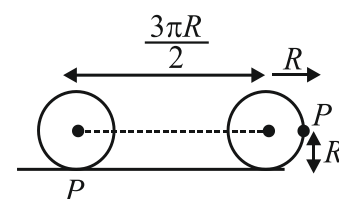
$$6 - 1.5t' = -6$$

$$12 = 1.5t'$$

$$t' = 12/1.5 = 8 \text{ s}.$$

(NCERT Ed. 2025-26, 11th, Page No. 30)

37. (2)



In three fourth revolution the center of wheel

$$\text{moves forward by } \frac{3}{4} \times 2\pi R = \frac{3\pi R}{2}$$

$$\text{Displacement} = \sqrt{R^2 + \left(\frac{3\pi R}{2} + R\right)^2}$$

$$= \sqrt{R^2 + \frac{R^2}{4}(3\pi + 2)^2}$$

$$= \frac{R}{2} \sqrt{4 + (3\pi + 2)^2}$$

(NCERT Ed. 2025-26, 11th, Page No. 33)

38. (4)

$$|\vec{a}_{avg}| = \frac{|\Delta \vec{v}|}{\Delta t}$$

$$= \frac{2v}{\frac{7}{2} \times \frac{2\pi r}{v}} = \frac{2v^2}{7\pi r}$$

(NCERT Ed. 2025-26, 11th, Page No. 42)

39. (1)

$$x = 4(1 - e^{-2t})$$

$$v = \frac{dx}{dt} = 4(0 - e^{-2t}(-2))$$

$$v = 4(1 + 2e^{-2t})$$

At $t = 0$

$$v = 8 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 14)

40. (3)

Given, $t = mx^2 + nx$

$$\frac{dt}{dx} = 2mx + n \Rightarrow \frac{dx}{dt} = \frac{1}{2mx + n}$$

$$v = \frac{1}{2mx + n}$$

Retardation of the motion.

$$a = v \frac{dv}{dx} = v \frac{-2m}{(2mx + n)^2} = -2mv^3$$

(NCERT Ed. 2025-26, 11th, Page No. 16)

41. (2)

When both velocity vectors make 45° with the horizontal the velocity vectors will become perpendicular.

Horizontal component remains same.

At time t , vertical component $= gt$

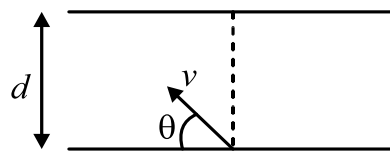
$$\tan 45^\circ = \frac{gt}{u} = 1$$

$$t = \frac{u}{g} = \frac{20}{10} = 2 \text{ s}$$

(NCERT Ed. 2025-26, 11th, Page No. 39)

42. (1)

$$t = \frac{d}{v \sin \theta}$$



For time to be minimum, $v \sin \theta$ should be maximum.

So, $\sin \theta = 1 \Rightarrow \theta = 90^\circ$

- River flow will not help the swimmer to cross the river.

(NCERT Ed. 2025-26, 11th, Page No. 31)

43. (2)

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

When $\theta = 0^\circ$, $\vec{A} \cdot \vec{B} = 8$

$\theta = 90^\circ$, $\vec{A} \cdot \vec{B} = 0$

When $\theta = 60^\circ$, $\vec{A} \cdot \vec{B} = 4$

When $\theta = 180^\circ$, $\vec{A} \cdot \vec{B} = -8$

(NCERT Ed. 2025-26, 11th, Page No. 32)

44. (1)

If the object is moving in negative direction the position time graph will have negative slope.

When speed reduces the position still increases so reason is wrong.

(NCERT Ed. 2025-26, 11th, Page No. 14)

45. (4)

$$\text{Line: } v^3 = 24 + (\tan 37^\circ)x^2$$

$$\text{Differentiate w.r.t. } x: 3v^2 \frac{dv}{dx} = 2(\tan 37^\circ)x$$

$$\Rightarrow a = v \frac{dv}{dx} = \frac{2(\tan 37^\circ)x}{3v}$$

$$\text{At } x = 2; v^3 = 24 + (\tan 37^\circ).4 = 27$$

$$\Rightarrow v = 3 \text{ m s}^{-1}$$

$$\text{So } a = \frac{2\left(\frac{3}{4}\right).2}{3.3} = \frac{1}{3} \text{ m s}^{-2}$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

46. (4)

$$n_1 = 1$$

$$n_2 = 4 \text{ (3rd excited state)}$$

∴ Number of spectrum lines

$$= \frac{n(n-1)}{2} = \frac{4 \times 3}{2} = 6$$

(NCERT Ed. 2025-26 11th Part-I Page No. 46)

47. (1)

- P₂O₅: Acidic oxide
- BeO: Amphoteric oxide
- Na₂O: Basic oxide
- NO: Neutral oxide

(NCERT Ed. 2025-26 11th Part-I Page No. 94)

48. (3)

Of all the elements, Hydrogen atom has the simplest line spectrum because hydrogen atom has only one electron and one proton.

(NCERT Ed. 2025-26 11th Part-I Page No. 45)

49. (3)

As per Moseley., a plot of $\sqrt{\nu}$ against atomic number (Z) is a straight line.

(NCERT Ed. 2025-26 11th Part-I Page No. 78)

50. (1)

- The energy of the electron in a hydrogen atom is lower than the energy of a free electron at rest.
- Angular momentum of electron in H-atom is quantised.

(NCERT Ed. 2025-26 11th Part-I Page No. 47)

51. (4)

Ge, As, Sb and Te, all are metalloids.

(NCERT Ed. 2025-26 11th Part-I Page No. 79)

52. (1)

The magnitude of velocity of electron in one electron species with increase of positive charge on the nucleus and principal quantum number increases and decreases respectively.

(NCERT Ed. 2025-26 11th Part-I Page No. 48)

53. (1)

$$(-\Delta_{\text{eg}}H): \text{S} > \text{Se} > \text{Te} > \text{O}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 90)

54. (2)

$$\Delta x = \frac{h}{4\pi m \Delta v}$$

$$= \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 20 \times 10^{-3} \times 33.1 \times \left(\frac{1}{100}\right)} = 7.96 \times 10^{-33} \text{ m}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 51)

55. (4)

Eu (Z = 63) belongs to lanthanoid series (3rd group and 6th period)

(NCERT Ed. 2025-26 11th Part-I Page No. 79)

56. (3)

$$\frac{r_3}{r_1} = \frac{(3)^2}{(1)^2} \quad \therefore r_3 = 9a_0$$

$$2\pi r = n\lambda$$

$$\lambda_3 = \frac{2\pi r_3}{n} = \frac{2\pi \times 9a_0}{3} = \frac{18}{3} \pi a_0$$

(NCERT Ed. 2025-26 11th Part-I Page No. 50, 48)

57. (3)

Metals are usually solids at room temperature [Mercury is an exception], Gallium and Caesium also have very low melting points (303 K and 302 K, respectively).

(NCERT Ed. 2025-26 11th Part-I Page No. 85)

58. (2)

Photoelectric effect can be explained by particle nature.

(NCERT Ed. 2025-26 11th Part-I Page No. 41)

59. (4)

Z = 105: Unnilpentium.

(NCERT Ed. 2025-26 11th Part-I Page No. 80)

60. (3)

v: γ -rays > X-rays > UV-rays > IR-rays > Microwaves > Radiowaves > Long radio waves

(NCERT Ed. 2025-26 11th Part-I Page No. 38)

61. (3)

Be and Al show diagonal relation.

(NCERT Ed. 2025-26 11th Part-I Page No. 93)

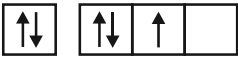
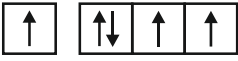
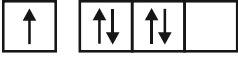
62. (3)

Chemical properties of atoms are controlled by the number of electrons, which are determined by the number of protons in the nucleus. Number of neutrons present in the nucleus have little effect on the chemical properties of an element. Therefore, all the isotopes of a given element show same chemical behaviour.

(NCERT Ed. 2025-26 11th Part-I Page No. 35)

63. (4)
For isoelectronic species, more the positive charge, lesser will be ionic radius.
 $\therefore$ ionic radius: $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
(NCERT Ed. 2025-26 11th Part-I Page No. 87)
64. (2)
Cathode ray behaviour can be observed with the help of certain kind of materials (phosphorescent or fluorescent) which glow when hit by them.
(NCERT Ed. 2025-26 11th Part-I Page No. 30)
65. (1)
IE: $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{O} < \text{N} < \text{F}$
(NCERT Ed. 2025-26 11th Part-I Page No. 88)
66. (3)
 $q = ne$
$$n = \frac{2.5 \times 10^{-16}}{1.602176 \times 10^{-19}} = 1560$$

(NCERT Ed. 2025-26 11th Part-I Page No. 71)
67. (2)
On Pauling scale, electronegativity of N and Cl both are 3.0.
(NCERT Ed. 2025-26 11th Part-I Page No. 91)
68. (1)
Number of neutrons = $A - Z$
 $\therefore A - Z = Z + \frac{Z \times 31.7}{100}$
 $A = 2Z + 0.317 Z$
 $81 = 2.317 Z$
 $Z = 34.958 \approx 35$
(NCERT Ed. 2025-26 11th Part-I Page No. 71)
69. (4)
 - Because of high reactivity, s-block elements are never found pure in nature.
 - d-block elements mostly form coloured ions.
(NCERT Ed. 2025-26 11th Part-I Page No. 84)
70. (1)
 - More the value of $(n + \ell)$, more will be energy of orbital
 - If $(n + \ell)$ value is same, more the value of n , more will be energy of orbital
 - $A: n + \ell = 4 + 2 = 6$
 $B: n + \ell = 3 + 2 = 5$
 $C: n + \ell = 4 + 1 = 5$
 $D: n + \ell = 3 + 2 = 5$ $\therefore$ Energy: $A > C > B = D$
(NCERT Ed. 2025-26 11th Part-I Page No. 61)

71. (3)
The valence of group-17 elements are 1 and 7.
(NCERT Ed. 2025-26 11th Part-I Page No. 92)
72. (3)
 - $r_n = 0.53 \frac{n^2}{Z} \text{ \AA}$
 - $v_n = 2.188 \times 10^6 \times \frac{Z}{n} \text{ ms}^{-1}$
 - $\text{KE} = 13.6 \frac{Z^2}{n^2} \text{ eV}$
 - Time period = $\frac{2\pi r}{v} \propto \frac{\left(\frac{n^2}{Z}\right)}{\left(\frac{Z}{n}\right)} \propto \frac{n^3}{Z^2}$
(NCERT Ed. 2025-26 11th Part-I Page No. 46, 47)
73. (1)
 $X: [\text{Uuo}]8s^1$
Valency of X is one so formula of compound will be XCl
(NCERT Ed. 2025-26 11th Part-I Page No. 79, 80)
74. (4)
Bohr's theory is applicable only for one electron species. eg: Be^{3+} , He^+ , H.
(NCERT Ed. 2025-26 11th Part-I Page No. 48)
75. (3)
 16^{th} group elements are known as chalcogen.
Chalcogens: O, S, Se, Te, Po, Lv
(NCERT Ed. 2025-26 11th Part-I Page No. 79, 84)
76. (4)
 -  : Aufbau rule is followed
 -  : Hunds rule is followed
 -  : Both Aufbau rule and Hunds rule are followed.
 -  : Both Aufbau rule and Hunds rule are violated.
(NCERT Ed. 2025-26 11th Part-I Page No. 62, 62)
77. (4)
As per modern periodic law, the physical and chemical properties of the elements are periodic functions of their atomic numbers.
(NCERT Ed. 2025-26 11th Part-I Page No. 78)

78. (2)

$$\bar{v} = \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{8}{9} \times R = R(1)^2 \left(\frac{1}{1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{8}{9} = \left(1 - \frac{1}{n_2^2} \right) \Rightarrow 1 - \frac{1}{3^2} = 1 - \frac{1}{n_2^2}$$

$$\therefore n_2 = 3$$

(NCERT Ed. 2025-26 11th Part-I Page No. 45)

79. (3)

- Eka-aluminium is Gallium.
- Predicted atomic weight of Eka-aluminium is 68.

(NCERT Ed. 2025-26 11th Part-I Page No. 76)

80. (3)

$$\lambda = 100 \text{ nm}$$

$$\therefore \lambda = \frac{h}{mv} = \frac{h}{m \times \left(\frac{\lambda}{100} \right)} = \frac{100 h}{m \lambda}$$

$$\therefore \lambda^2 = \frac{100 h}{m}$$

$$\therefore \lambda = 10 \sqrt{\frac{h}{m}}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 50)

81. (3)

Lothar Meyer plotted the physical properties such as melting point and boiling point against atomic weight.

(NCERT Ed. 2025-26 11th Part-I Page No. 75)

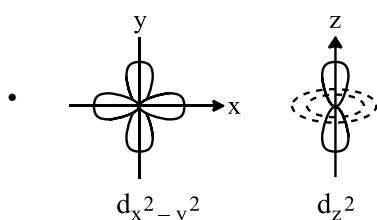
82. (1)

$$\text{Radial nodes} = n - \ell - 1$$

- $5s : 5 - 0 - 1 = 4$
- $5p_y : 5 - 1 - 1 = 3$
- $4d_{yz} : 4 - 2 - 1 = 1$
- $5d_{x^2-y^2} : 5 - 2 - 1 = 2$

(NCERT Ed. 2025-26 11th Part-I Page No. 59)

83. (4)



- All 3d orbitals are equivalent in energy

(NCERT Ed. 2025-26 11th Part-I Page No. 59)

84. (4)

$$\text{Orbital angular momentum} = \sqrt{\ell(\ell+1)} \frac{h}{2\pi}$$

- For s-orbital: $\sqrt{0(0+1)} \frac{h}{2\pi} = 0$
- For p-orbital: $\sqrt{1(1+1)} \frac{h}{2\pi} = \frac{h}{\sqrt{2}\pi}$
- For d-orbital: $\sqrt{2(2+1)} \frac{h}{2\pi} = \sqrt{\frac{3}{2}} \frac{h}{\pi}$

(NCERT Ed. 2025-26 11th Part-I Page No. 55)

85. (2)

Maximum number of electrons present in any orbit
 $= 2(2\ell + 1)$

(NCERT Ed. 2025-26 11th Part-I Page No. 55)

86. (3)

$$h\nu = h\nu_0 + \text{KE}$$

$$h\nu = h\nu_0 + 0$$

$$\therefore \nu_0 = \nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{3000 \times 10^{-10}} \text{ s}^{-1}$$

$$\nu_0 = 10^{15} \text{ s}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 42)

87. (3)

Group-18 (inert element) contain non-metals only.

(NCERT Ed. 2025-26 11th Part-I Page No. 79)

88. (4)

Heisenberg uncertainty principle is a direct consequence of wave-particle duality and significant only for microscopic objects, which can be represented as:

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$$\text{or } \Delta E \times \Delta t \geq \frac{h}{4\pi}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 51)

89. (1)

5th period: Rb(Z = 37) to Xe(Z = 54)

6th period: Cs(Z = 55) to Rn(Z = 86)

(NCERT Ed. 2025-26 11th Part-I Page No. 79)

90. (3)

The ground state electronic configuration of Cr-atom is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$ because the ground state electronic configuration of the atom of an element always corresponds to the state of the lowest total electronic energy.

(NCERT Ed. 2025-26 11th Part-I Page No. 64)

91. (2)

All organisms, even the largest, start their life from a single cell, called zygote. You may wonder how a single cell then goes on to form such large organisms. Growth and reproduction are characteristics of cells, indeed of all living organisms. All cells reproduce by dividing into two, with each parental cell giving rise to two daughter cells each time they divide. These newly formed daughter cells can themselves grow and divide, giving rise to a new cell population that is formed by the growth and division of a single parental cell and its progeny. In other words, such cycles of growth and division allow a single cell to form a structure consisting of millions of cells.

(NCERT Ed. 2025-26, 11th, Page No. 120)

92. (3)

Since the number of chromosomes in the parent and progeny cells is the same, mitosis is also called as equational division. At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells. While organelles are distributed to the two daughter cells during cytokinesis, the division is not precisely equal.

(NCERT Ed. 2025-26, 11th, Page No. 122)

93. (3)

M phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell. The M Phase starts with the nuclear division, corresponding to the separation of daughter chromosomes (karyokinesis) and usually ends with division of cytoplasm (cytokinesis). In the 24 hour average duration of cell cycle of a human cell, cell division proper lasts for only about an hour. The interphase lasts more than 95% of the duration of cell cycle. The cells that do not divide further exit G₁ phase to enter an inactive stage called quiescent stage (G₀) of the cell cycle.

(NCERT Ed. 2025-26, 11th, Page No. 122)

94. (3)

The stage between the two meiotic divisions is called interkinesis and is generally short lived. There is no replication of DNA during interkinesis. Interkinesis is followed by prophase II, a much simpler prophase than prophase I.

(NCERT Ed. 2025-26, 11th, Page No. 127)

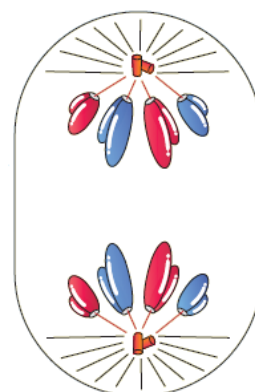
95. (1)

During zygotene, chromosomes start pairing together and this process of association is called synapsis. Pachytene is characterised by the appearance of recombination nodules, the sites at which crossing over occurs. The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers. These X-shaped structures, are called chiasmata. The final stage of meiotic prophase I is diakinesis. This is marked by terminalisation of chiasmata.

(NCERT Ed. 2025-26, 11th, Page No. 126)

96. (2)

The diagram illustrates the anaphase stage of mitosis. Chromosome decondensation (uncoiling) occurs during telophase, the stage that follows anaphase. In anaphase, the chromosomes are still in their highly condensed form. At the onset of anaphase, each chromosome arranged at the metaphase plate is split simultaneously and the two daughter chromatids, now referred to as daughter chromosomes (B is one of them) of the future daughter nuclei, begin their migration towards the two opposite poles. 'C' points to the centrosome at one of the cell's poles. The equatorial plate (or metaphase plate) is the imaginary plane in the middle of the cell where chromosomes align during Metaphase. The spindle fibers originating from the centrosomes are responsible for this alignment, but the centrosome itself does not form the plate. The movement of the daughter chromosomes towards the poles is driven by the shortening of the microtubules (like A), not their elongation.



Anaphase

(NCERT Ed. 2025-26, 11th, Page No. 123)

97. (3)
In onion root tip ($2n$) cells, there are 16 chromosomes in each cell. Each of the gametes in onion will thus have 8 cells.
(NCERT Ed. 2025-26, 11th, Page No. 122)
98. (4)
Centriole movement towards opposite poles is a characteristic event in prophase of mitosis (and prophase II of meiosis). A somatic cell does not undergo meiosis, so in it bivalents will not be formed. DNA replication occurs during the S phase of interphase, which precedes mitosis. If the centrioles are moving, the cell is already in prophase, meaning DNA replication has already been completed. After DNA replication in S phase, each chromosome consists of two identical sister chromatids, not four.
(NCERT Ed. 2025-26, 11th, Page No. 122)
99. (4)
In zygotene, homologous chromosomes pair up (synapsis) to form bivalents (or tetrads of chromatids). A tetrad of cells is formed by cytokinesis after telophase II.
(NCERT Ed. 2025-26, 11th, Page No. 126)
100. (1)
Formation of mitotic apparatus: This happens at the beginning of Meiosis II (Prophase II). The spindle fibers start to form again.
Attachment of spindle fibres to kinetochores of chromosomes: This occurs in Prophase II/early Metaphase II, as the chromosomes move towards the metaphase plate.
Alignment of chromosomes on equator: This is characteristic of Metaphase II, where the chromosomes (each still composed of two sister chromatids) line up at the equatorial plate.
Formation of daughter chromosomes: This happens during Anaphase II, when the sister chromatids separate and move to opposite poles, becoming individual daughter chromosomes.
Division of cytoplasm: This is cytokinesis, which typically follows telophase II, resulting in four haploid daughter cells.
(NCERT Ed. 2025-26, 11th, Page No. 127)
101. (3)
Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.
(NCERT Ed. 2025-26, 11th, Page No. 122)
102. (1)
The interphase lasts more than 95% of the duration of cell cycle.
(NCERT Ed. 2025-26, 11th, Page No. 121)
103. (3)
Bivalents are formed during Prophase I of meiosis. A bivalent consists of two homologous chromosomes, each with two sister chromatids, paired up. This pairing is crucial for crossing over and proper segregation. The number of bivalents is equal to the haploid number (n) of chromosomes.
(NCERT Ed. 2025-26, 11th, Page No. 126)
104. (2)
Telophase shows the following key events: Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements. Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei. Nucleolus, golgi complex and ER reform. Centromere splitting occurs in Anaphase.
(NCERT Ed. 2025-26, 11th, Page No. 124)
105. (2)
Interphase: Interphase is divided into three sub-stages: G_1 , S, and G_2 .
Karyokinesis: Karyokinesis refers to nuclear division, which typically involves four main phases: Prophase, Metaphase, Anaphase, and Telophase (in mitosis and meiosis II).
Prophase I: Prophase I of meiosis is a complex and extended phase that is further divided into five sub-stages: Leptotene, Zygotene, Pachytene, Diplotene, and Diakinesis.
Cell cycle: The cell cycle broadly consists of two major phases: Interphase and M-phase (mitosis/meiosis and cytokinesis).
(NCERT Ed. 2025-26, 11th, Page No. 121)
106. (2)
In plant cells, wall formation starts in the centre of the cell and grows outward to meet the existing lateral walls. The formation of the new cell wall begins with the formation of a simple precursor, called the cell-plate that represents the middle lamella between the walls of two adjacent cells.
(NCERT Ed. 2025-26, 11th, Page No. 124)

107. (1)

Prophase which is the first stage of karyokinesis of mitosis follows the S and G₂ phases of interphase. In the S and G₂ phases, the new DNA molecules formed are not distinct but intertwined. Prophase is marked by the initiation of condensation of chromosomal material. The chromosomal material becomes untangled during the process of chromatin condensation. Cells at the end of prophase, when viewed under the microscope, do not show golgi complexes, endoplasmic reticulum, nucleolus and the nuclear envelope. Prokaryotic cells do not undergo mitosis as they do not have well-defined nucleus.

(NCERT Ed. 2025-26, 11th, Page No. 122)

108. (1)

A very significant contribution of mitosis is cell repair. Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of higher plants throughout their life.

(NCERT Ed. 2025-26, 11th, Page No. 125)

109. (3)

The cells of the upper layer of the epidermis, cells of the lining of the gut, and blood cells are being constantly replaced due to mitosis in certain cells. If mitosis stops in all the cells of a person, the lining of the gut, would soon dissolve, leading to severe digestive issues and other health problems. The gamete formation and introduction of variation in progeny occurs by meiosis in humans. The dead cells are replaced by new living cells produced by mitosis.

(NCERT Ed. 2025-26, 11th, Page No. 125)

110. (2)

In animals, mitotic cell division is only seen in the diploid somatic cells. However, there are few exceptions to this where haploid cells divide by mitosis, for example, male honey bees.

(NCERT Ed. 2025-26, 11th, Page No. 122)

111. (3)

In oocytes of some vertebrates, diplotene can last for months or years.

(NCERT Ed. 2025-26, 11th, Page No. 126)

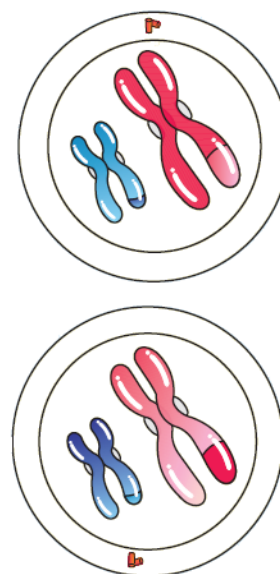
112. (1)

The final stage of meiotic prophase I is diakinesis. This is marked by terminalisation of chiasmata. During this phase the chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosomes for separation.

(NCERT Ed. 2025-26, 11th, Page No. 126)

113. (3)

The given image represents telophase I. 'A' is the kinetochore; the site of attachment of microtubules from the spindle fibres. 'B' is the site of crossing over. 'C' is the envelope that surrounds the segregated homologous chromosomes. 'D' does not disintegrate, rather prepares for mitosis II.



Telophase I

(NCERT Ed. 2025-26, 11th, Page No. 127)

114. (4)

DNA replication (synthesis) occurs during the S phase, but it takes place in the nucleus, not the cytoplasm. Synapsis, the pairing of homologous chromosomes, occurs during zygotene, which precedes diplotene. In diplotene, the homologous chromosomes begin to separate, but remain attached at chiasmata. Meiosis I is the reduction division, where the chromosome number is halved. Meiosis II is an equational division, similar to mitosis, where sister chromatids separate. In Anaphase II of meiosis, the centromeres of each chromosome divide, allowing the sister chromatids to separate and move to opposite poles of the cell.

(NCERT Ed. 2025-26, 11th, Page No. 127)

115. (3)

Cell furrow forms during cytokinesis in animal cells.

(NCERT Ed. 2025-26, 11th, Page No. 124)

116. (3)

Telophase I: The nuclear membrane and nucleolus reappear, cytokinesis follows and this is called as dyad of cells. Although in many cases the chromosomes do undergo some dispersion, they do not reach the extremely extended state of the interphase nucleus. The stage between the two meiotic divisions is called interkinesis and is generally short lived.

(NCERT Ed. 2025-26, 11th, Page No. 127)

117. (3)

At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells.

(NCERT Ed. 2025-26, 11th, Page No. 124)

118. (2)

The replicated chromosomes (DNA) are then distributed to daughter nuclei by a complex series of events during cell division.

(NCERT Ed. 2025-26, 11th, Page No. 120)

119. (3)

Interphase: This is the longest phase of the cell cycle, where the cell grows, synthesizes DNA, and prepares for division. It's significantly longer than any of the meiotic phases.

Meiosis I: This is the first meiotic division and includes Prophase I, Metaphase I, Anaphase I, and Telophase I. Prophase I itself is very long, and the entire Meiosis I takes a considerable amount of time.

Prophase I: While a part of Meiosis I, Prophase I is the longest stage within Meiosis I due to the complex processes of chromosome condensation, synapsis, and crossing over.

Prophase II: This is the prophase of the second meiotic division. It is much shorter than Prophase I as there's no DNA replication or crossing over, and chromosomes are already condensed.

(NCERT Ed. 2025-26, 11th, Page No. 126)

120. (2)

Meiosis is the mechanism by which conservation of specific chromosome number of each species is achieved across generations in sexually reproducing organisms. Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms whereas fertilisation restores the diploid phase.

(NCERT Ed. 2025-26, 11th, Page No. 128)

121. (1)

Chromosomes are composed of two chromatids attached together at the centromere. Each centrosome radiates out microtubules called asters. The two asters together with spindle fibres forms mitotic apparatus. The plane of alignment of the chromosomes at metaphase is referred to as the metaphase plate.

(NCERT Ed. 2025-26, 11th, Page No. 122)

122. (2)

The daughter cells formed after meiosis during spermatogenesis in human males are equal in size. The daughter cells formed after meiosis during oogenesis in human females are unequal in size.

(NCERT Ed. 2025-26, 11th, Page No. 130)

123. (2)

The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers. These X-shaped structures, are called chiasmata.

(NCERT Ed. 2025-26, 11th, Page No. 126)

124. (1)

Cells at the end of prophase, when viewed under the microscope, do not show golgi complexes, endoplasmic reticulum, nucleolus and the nuclear envelope. By the end of prophase, the chromosomes are highly condensed and clearly visible. Daughter chromosomes move towards opposite poles during anaphase.

(NCERT Ed. 2025-26, 11th, Page No. 123)

125. (3)

In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium (e.g., liquid endosperm in coconut).

(NCERT Ed. 2025-26, 11th, Page No. 124)

126. (3)

Plant cells, are enclosed by a relatively inextensible cell wall, therefore they undergo cytokinesis by a different mechanism than animal cells. In plant cells, wall formation starts in the centre of the cell.

(NCERT Ed. 2025-26, 11th, Page No. 124)

127. (4)

G₁ phase corresponds to the interval between mitosis and initiation of DNA replication.

(NCERT Ed. 2025-26, 11th, Page No. 121)

128. (2)

The final stage of meiotic prophase I is diakinesis. Diakinesis represents transition to metaphase.

(NCERT Ed. 2025-26, 11th, Page No. 126)

129. (1)

A daughter chromosome consists of only a single chromatid.

(NCERT Ed. 2025-26, 11th, Page No. 123)

130. (4)

Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G₁ phase to enter an inactive stage called quiescent stage (G₀) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

(NCERT Ed. 2025-26, 11th, Page No. 122)

131. (1)

In the first meiotic division the homologous chromosomes pair to form bivalents, and undergo crossing over during prophase I. During metaphase I the bivalents arrange on the equatorial plate. This is followed by anaphase I in which separated homologous chromosomes move to the opposite poles with both their chromatids. Each pole receives half the chromosome number of the parent cell. Meiosis II is similar to mitosis. During anaphase II the sister chromatids separate. Thus at the end of meiosis (cytokinesis after telophase II) four haploid cells are formed.

(NCERT Ed. 2025-26, 11th, Page No. 126)

132. (3)

Given that the gamete has 20 pg of DNA, this represents the DNA content of a haploid cell (c). Therefore, a diploid parent cell in the G₁ phase (before DNA replication) would have double this amount, which is 40 pg (2c). After S phase, in the G₂ phase, the DNA amount is double i.e. 80 pg. In Metaphase I of meiosis, the cell is still in its replicated state, with homologous chromosomes paired up. Therefore, the DNA content remains the same as in the G₂ phase, which is 80 pg. Meiosis I concludes with the separation of homologous chromosomes into two daughter cells. Each of these cells is now haploid in terms of chromosome number, but each chromosome still consists of two sister chromatids. Thus, the DNA content is halved from the G₂/Metaphase I stage. This results in 40 pg of DNA in each cell during Prophase II.

(NCERT Ed. 2025-26, 11th, Page No. 121)

133. (4)

The "resting phase" of the cell cycle is known as interphase. During interphase, the cell actively grows and increases in size to prepare for division. The cell produces more organelles during the G₁ phase to ensure that the daughter cells will be properly equipped after division. The amount of DNA is doubled, but the chromosome number remains the same.

(NCERT Ed. 2025-26, 11th, Page No. 121)

134. (1)

In animals, mitotic cell division is only seen in the diploid somatic cells. However, there are few exceptions to this where haploid cells divide by mitosis, for example, male honey bees. Against this, the plants can show mitotic divisions in both haploid and diploid cells.

(NCERT Ed. 2025-26, 11th, Page No. 125)

135. (2)

In anaphase I, separated homologous chromosomes move to the opposite poles with both their chromatids. Each pole receives half the chromosome number of the parent cell.

(NCERT Ed. 2025-26, 11th, Page No. 129)

136. (3)

The given bond is a peptide (amide) bond —CO—NH— found between amino acids in polypeptides/proteins (including protein enzymes). Fatty acids contain a carboxyl group and a hydrocarbon chain, not peptide bonds.

(NCERT Ed. 2025-26, 11th, Page No. 107)

137. (3)

List-I (Category)	List-II (Secondary Metabolites)
Toxins	Abrin, Ricin
Terpenoids	Monoterpenes, Diterpenes
Alkaloids	Morphine, Codeine
Lectins	Concanavalin A

(NCERT Ed. 2025-26, 11th, Page No. 108)

138. (3)

Enzymes are highly specific due to the unique shape of their active site, which fits a particular substrate like a key in a lock. This is known as enzyme-substrate specificity. For example, the enzyme sucrase will only break down the substrate sucrose into glucose and fructose, and will not act on other sugars like lactose or maltose.

A single type of substrate molecule can be acted upon by multiple enzymes, but not at the same time. This occurs sequentially in metabolic pathways. The product of one enzyme-catalyzed reaction becomes the substrate for the next enzyme in the pathway. For instance, in glycolysis, a single glucose molecule undergoes a series of transformations, where different enzymes act on it at various stages to eventually convert it into pyruvate.

(NCERT Ed. 2025-26, 11th, Page No. 114)

139. (1)

Unsaturated fats are more chemically reactive than saturated fats. The presence of carbon-carbon double bonds ($\text{C}=\text{C}$) in their structure makes them more susceptible to addition reactions, such as hydrogenation, where hydrogen atoms are added across the double bonds.

The reason for the increased reactivity of unsaturated fats is the presence of one or more $\text{C}=\text{C}$ double bonds, not their absence. Saturated fats have only single bonds (C—C).

(NCERT Ed. 2025-26, 11th, Page No. 106)

140. (1)

- The human body can synthesize some amino acids, but it cannot synthesize all of them. The amino acids that the body cannot produce are called essential amino acids, and they must be obtained through the diet. This is why a consistent dietary intake of protein is essential for health.
- Valine and leucine are two examples of essential amino acids that the body cannot synthesize and must be acquired from food.
- Aspartic acid and glutamic acid are considered acidic amino acids because their R-groups contain a second carboxyl (—COOH) group. At physiological pH, this group is deprotonated, giving the amino acid a net negative charge.
- Lysine and arginine are classified as basic amino acids because their R-groups contain an amino (—NH_2) group that accepts a proton, giving the amino acid a net positive charge at physiological pH.

(NCERT Ed. 2025-26, 11th, Page No. 109)

141. (2)

- Serine: The side chain is $\text{—CH}_2\text{OH}$. This is characteristic of the amino acid serine.
- Cysteine: The side chain is $\text{—CH}_2\text{SH}$. The presence of a sulfhydryl group (SH) identifies it as cysteine.
- Tyrosine: The side chain is —CH_2 followed by a benzene ring with a hydroxyl group. This is the structure of tyrosine.
- Glutamic acid: The side chain is $\text{—CH}_2\text{CH}_2\text{COOH}$. The presence of an extra carboxyl group (COOH) makes this glutamic acid, an acidic amino acid.

(NCERT Ed. 2025-26, 11th, Page No. 107)

142. (3)

- Melting of ice is a physical change, not a chemical change. A physical change alters the form or appearance of a substance but does not change its chemical composition. Ice (H_2O) changes from a solid to a liquid, but it remains water (H_2O).
- RuBisCo (Ribulose-1,5-bisphosphate carboxylase/oxygenase) is the most abundant protein in the world because it is an essential enzyme in photosynthesis, the process by which plants produce their food.
- Collagen is the most abundant protein in the animal kingdom, serving as a primary structural protein in connective tissues like skin, bones, and cartilage.
- A chemical reaction involves the breaking of old chemical bonds and the formation of new ones to create new substances.

(NCERT Ed. 2025-26, 11th, Page No. 113)

143. (2)

$\text{Ba(OH)}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{H}_2\text{O}$ is a classic example of a neutralization reaction, where a strong acid (H_2SO_4) reacts with a strong base (Ba(OH)_2) to form a salt (BaSO_4) and water (H_2O).

Enzymes are highly efficient biological catalysts. They work by lowering the activation energy of a reaction, which significantly increases the reaction rate, often by a factor of millions or billions.

A metabolic pathway is a series of chemical reactions where each step is catalyzed by a different, specific enzyme. A complex of enzymes might catalyze a series of steps, but it is not typically the same single enzyme. The products of one enzyme-catalyzed reaction serve as the substrate for the next enzyme in the pathway.

Thermal stability is one of the most important quality of thermostable enzymes isolated from thermophilic organisms.

(NCERT Ed. 2025-26, 11th, Page No. 113)

144. (3)

Rate of a physical or chemical process refers to the amount of product formed per unit time.

$$\text{rate} = \frac{\delta P}{\delta t}$$

Given in question:

Product (δP): The amount of product formed is 600,000 molecules of CO_2 .

Time (δt): The time taken is 1 minute.

Rate of this enzymatic reaction, expressed in molecules of product formed per second is;

- Rate = (Total Product) / (Total Time)
- Rate = 600,000 molecules / 60 seconds
- Rate = 10,000 molecules/second

(NCERT Ed. 2025-26, 11th, Page No. 114)

145. (2)

Primary metabolites are defined as compounds directly involved in the normal growth, development, and reproduction of an organism.

Example: Glucose, fatty acids and amino acids

As there are essential amino acids that the body can't produce, certain fatty acids, such as omega-3 and omega-6 fatty acids, are essential and must be obtained from the diet. The body can synthesize other fatty acids, which are considered non-essential.

(NCERT Ed. 2025-26, 11th, Page No. 108)

146. (4)

List-I	List-II
Cholesterol	A component of animal cell membrane
Adenylic acid	Pentose sugar + Nitrogenous base + Phosphate group
Adenosine	Pentose sugar + Nitrogenous base
Proline	Non-essential amino acid

(NCERT Ed. 2025-26, 11th, Page No. 106)

147. (3)

Enzymes possess a specific region called the active site, which is a three-dimensional pocket or cleft where the substrate binds. The substrate has a corresponding reactive site, which is the part of the molecule that undergoes the chemical transformation during the reaction. The specificity of the enzyme-catalyzed reaction depends on the complementary shapes and chemical properties of the active and reactive sites.

The purpose of the active and reactive sites is to facilitate binding and catalysis, not repulsion. The enzyme's active site and the substrate's reactive site are designed to attract and interact with each other. This interaction is essential for the enzyme to lower the activation energy and speed up the chemical reaction. Repulsion would prevent the enzyme from functioning.

(NCERT Ed. 2025-26, 11th, Page No. 115)

148. (2)

- Glycolysis, the breakdown of glucose, produces pyruvic acid (pyruvate) in the cytoplasm of virtually all living cells. Under aerobic conditions, this pyruvate then enters the mitochondria for the next stage of cellular respiration.
- During intense exercise, oxygen supply to muscle cells becomes limited. To allow glycolysis to continue and produce ATP, pyruvate is converted to lactic acid.
- Yeast and some bacteria perform alcoholic fermentation, producing ethanol and CO_2 from pyruvate. Human muscle cells perform lactic acid fermentation, not alcoholic fermentation.
- The final product of the glycolytic pathway is pyruvate. However, the metabolic fate of that pyruvate is what changes based on the organism and oxygen availability. Pyruvate can be converted to lactate in human muscle or to ethanol and CO_2 in yeast.

(NCERT Ed. 2025-26, 11th, Page No. 114)

149. (4)

- Glucose: An aldohexose. In its open-chain form, it has a terminal aldehyde group ($-\text{CHO}$), which contains a carbonyl.
- Fructose: A ketohexose. In its open-chain form, it has a ketone group ($>\text{C}=\text{O}$) within the carbon chain.
- Pyruvic Acid (Pyruvate): The end product of glycolysis. Its structure is $\text{CH}_3-\text{C}(=\text{O})-\text{COOH}$. It clearly contains a ketone group ($>\text{C}=\text{O}$).
- Amino Acid: The general structure is $\text{R}-\text{CH}(\text{NH}_2)-\text{COOH}$. The functional groups are an amino group and a carboxyl group ($-\text{COOH}$). While a carboxyl group contains a $\text{C}=\text{O}$ double bond, it behaves chemically very differently from an aldehyde or ketone. Standard tests for carbonyl groups are typically designed for aldehydes and ketones.
- Fatty Acid: A long hydrocarbon chain with a terminal carboxyl group ($-\text{COOH}$).
- Phospholipid: A complex lipid. It has ester linkages but does not have a free aldehyde or ketone group.
- Nucleotide: Composed of a phosphate, a pentose sugar, and a nitrogenous base. The pentose sugar (ribose or deoxyribose) *does* have a potential aldehyde group, but the nitrogenous bases (purines and pyrimidines) also contain carbonyl groups in some cases (e.g., Guanine, Thymine, Cytosine, Uracil).

(NCERT Ed. 2025-26, 11th, Page No. 105)

150. (3)

- Water is a highly polar molecule and its molecules form strong hydrogen bonds with each other.
- Oil is primarily composed of triglycerides, which are a type of lipid. A triglyceride molecule consists of a glycerol backbone and three long fatty acid chains. These long chains are made of carbon and hydrogen ($-\text{CH}_2-\text{CH}_2-\text{CH}_2-$) and are extremely nonpolar.
- When oil and water are mixed, the nonpolar oil molecules cannot form hydrogen bonds with the polar water molecules. From an energy perspective, it is much more favorable for the water molecules to maintain their hydrogen-bonding network with each other. This results in the water molecules effectively "pushing away" or excluding the oil molecules. The excluded oil molecules then aggregate together (driven by hydrophobic interactions) to minimize their contact with water, forming a separate layer.

(NCERT Ed. 2025-26, 11th, Page No. 106)

151. (3)

The figure shows the different levels of protein structure.

- Primary structure is the linear sequence of amino acids in a polypeptide chain.
- Secondary structure is the local folding of the polypeptide chain into repeating structures like the alpha-helix (A) and beta-pleated sheets (B).
- The formation and stabilization of both the alpha-helix and beta-pleated sheets are driven by the formation of hydrogen bonds between the backbone atoms of the polypeptide chain. The hydrogen bonds form between the hydrogen atom of an amide group ($-\text{NH}$) and the oxygen atom of a carbonyl group ($-\text{C}=\text{O}$).

(NCERT Ed. 2025-26, 11th, Page No. 112)

152. (1)

Average Composition of Cells

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(NCERT Ed. 2025-26, 11th, Page No. 109)

153. (3)

All amino acids, except for glycine, have a central alpha-carbon bonded to four different groups: an amino group ($-\text{NH}_2$), a carboxyl group ($-\text{COOH}$), a hydrogen atom ($-\text{H}$), and a unique side chain ($-\text{R}$). In glycine, the side chain is also a hydrogen atom ($-\text{H}$). This means its alpha-carbon is bonded to two identical hydrogen atoms, making it symmetrical or achiral (not asymmetric).

Some competitive inhibitors are often used in the control of bacterial pathogens.

(NCERT Ed. 2025-26, 11th, Page No. 117)

154. (4)

- Cellulose is a polysaccharide made up of a long chain of only one type of monosaccharide, glucose.
- Inulin is a storage polysaccharide found in plants like onions and garlic, composed exclusively of repeating units of fructose.
- The helical structure of amylose, a component of starch, traps iodine molecules, resulting in a characteristic blue-black color.
- Nucleic acids (DNA and RNA) are polymers made up of different types of monomers, known as nucleotides. These nucleotides vary based on the nitrogenous base (A, T, C, G in DNA or A, U, C, G in RNA), making the overall polymer a heteropolymer.

(NCERT Ed. 2025-26, 11th, Page No. 110)

155. (1)

Sucrose is a non-reducing sugar. Reducing sugars are those that have a free aldehyde or ketone group that can donate electrons to other molecules. Benedict's and Tollens' reagents detect these free groups. Since sucrose does not react with them, it is classified as a non-reducing sugar.

The specific way sucrose is formed is what makes it a non-reducing sugar. The glycosidic bond is formed between the anomeric carbon (C-1) of glucose and the anomeric carbon (C-2) of fructose. An anomeric carbon is the carbon that was part of the aldehyde or ketone group in the open-chain form of the sugar. Because these key carbons are involved in the bond, they are "locked" and cannot open up to form the free aldehyde group needed to be a reducing agent. Thus, the formation of this specific linkage directly explains why sucrose cannot reduce Benedict's or Tollens' reagents.

(NCERT Ed. 2025-26, 11th, Page No. 110)

156. (3)

List-I	List-II
Ions	Form 1% of the total cellular mass
Lipids	Not strictly macromolecule
Lecithin	A component of cell membrane
DNA and RNA	Genetic material

(NCERT Ed. 2025-26, 11th, Page No. 109)

157. (3)

The acid-soluble pool (A) is the filtrate from a homogenized tissue treated with trichloroacetic acid. It contains small biomolecules with low molecular weights, such as monosaccharides, amino acids, and nucleotides.

- (i) Molecular weight ranging from 18 to 800 Daltons (Da) approximately. This characteristic belongs to the acid-soluble pool (A).
- (iv) Has monomers. Monomers are the building blocks of polymers and are small, so this characteristic belongs to the acid-soluble pool (A).

The acid-insoluble pool (B) is the retentate, or the fraction that doesn't pass through the filter. It contains large biomolecules (macromolecules) that have precipitated out.

- (ii) Proteins, nucleic acids, polysaccharides, and lipids. These are the major macromolecules and are found in the acid-insoluble pool (B). Note that lipids are an exception to the molecular weight rule, but they are still found

in the acid-insoluble fraction because their large, non-polar structure causes them to aggregate and not pass through the filter.

- (iii) Contain chemicals that have molecular weight more than 10000 Da and above. This is the defining characteristic of macromolecules and belongs to the acid-insoluble pool (B).
- (v) Generally, has polymers. Polymers are large molecules formed from monomers and are found in the acid-insoluble pool (B).

(NCERT Ed. 2025-26, 11th, Page No. 108)

158. (3)

Protein denaturation involves the unfolding of a protein's secondary and tertiary structures without breaking its primary structure.

- Hydrogen bonds are crucial for maintaining the secondary structures like alpha-helices and beta-sheets. Extreme heat or pH changes can disrupt these bonds, causing the protein to unfold.
- The three-dimensional folding (tertiary and quaternary structures) is a protein's unique shape essential for its function. Denaturation leads to the loss of this specific folding.

Hydrophobic interactions and ionic bonds are key forces that stabilize a protein's tertiary structure. These are broken during denaturation, allowing the protein to assume a random, non-functional shape. Hydrolysis of peptide bonds is the breakdown of the protein's primary structure into its constituent amino acids. This process, called digestion or proteolysis, is a different chemical reaction and does not occur during simple denaturation by heat or pH. Denaturation is a change in shape, while hydrolysis is a change in chemical composition.

(NCERT Ed. 2025-26, 11th, Page No. 112)

159. (2)

Purines are indeed heterocyclic, meaning they contain rings with more than one type of atom (carbon and nitrogen). However, they are composed of a double-ring structure. A purine has a total of four nitrogen atoms in its rings, not five.

Pyrimidines have a single-ring structure that is hexagonal (six-sided) and contains two nitrogen atoms, one at position 1 and the other at position 3.

(NCERT Ed. 2025-26, 11th, Page No. 107)

160. (3)

- Enzymes are biological catalysts that speed up chemical reactions in the body without being consumed in the process. They regulate the rate of metabolism.
- Vitamins are organic compounds that are essential for normal bodily functions. Many vitamins act as coenzymes or precursors to coenzymes, helping enzymes to function correctly in metabolic reactions.
- Hormones are chemical messengers that travel through the bloodstream to target cells, where they bind to receptors and initiate specific responses. They regulate a wide range of physiological processes, including metabolism, growth, and reproduction.

(NCERT Ed. 2025-26, 11th, Page No. 118)

161. (1)

Malonate is a competitive inhibitor of succinic dehydrogenase. Its chemical structure is very similar to that of the enzyme's natural substrate, succinate, which allows it to bind to the enzyme's active site and block succinate from binding. This is a classic example of competitive inhibition.

The reason describes non-competitive or allosteric inhibition, not the mechanism of malonate's action. Malonate is a competitive inhibitor and binds directly to the active site, not an allosteric site

(NCERT Ed. 2025-26, 11th, Page No. 117)

162. (3)

Holoenzyme is an enzyme with its cofactor, making it catalytically active. The protein part is the apoenzyme, which is inactive on its own.

Ribozyme is a catalytic RNA molecule, not a polypeptide chain. Its catalytic activity resides in its nucleic acid structure, not its protein part.

Co-enzyme - An organic, non-protein molecule that is transiently bound to an enzyme and is often derived from vitamins.

Prosthetic group is a non-protein component of an enzyme, but it is tightly and permanently bound, often covalently. Metal ions that are essential for enzyme function are called activators, and they can be loosely bound.

(NCERT Ed. 2025-26, 11th, Page No. 118)

163. (3)

Enzymes function as biological catalysts. They accelerate a reaction by lowering the activation energy—the minimum energy required for the reaction to occur.

The specific function of an enzyme is determined by its unique three-dimensional shape, particularly the shape of its active site. This is known as the "lock-and-key" or "induced fit" model, where only a specific substrate can bind to the enzyme's active site.

Enzymes do not change the overall free energy change (ΔG) of a reaction. They only affect the rate at which the reaction reaches equilibrium. They cannot make a non-spontaneous reaction (one with a positive ΔG) spontaneous.

In feedback inhibition, the final product of a metabolic pathway acts as an allosteric inhibitor, binding to an enzyme earlier in the pathway to inhibit its activity. This regulatory mechanism prevents the cell from wasting resources by overproducing the product.

(NCERT Ed. 2025-26, 11th, Page No. 115)

164. (3)

The reaction shown is the hydrolysis of maltose:

$$C_{12}H_{22}O_{11} + H_2O \rightarrow 2C_6H_{12}O_6$$

- Maltose is a disaccharide.
- Water (H_2O) is used to break the glycosidic bond.
- The product is glucose, a monosaccharide.

Enzyme A, which is maltase, catalyzes this reaction.

(NCERT Ed. 2025-26, 11th, Page No. 117)

165. (3)

Natural gums are complex carbohydrates (polysaccharides) found in plants.

Fevicol is a man-made substance (a synthetic polymer) with a different chemical composition, primarily made of polyvinyl acetate (PVA).

(NCERT Ed. 2025-26, 11th, Page No. 119)

166. (1)

The organic compound described is nucleic acids (DNA and RNA), which indeed have their monomeric units (nucleotides) joined by phosphodiester bonds.

When a dry living tissue is completely burnt, the carbon-based organic compounds are oxidized to form gases like carbon dioxide (CO_2) and water vapor. However, inorganic elements like Calcium (Ca), Magnesium (Mg), and Sulphur (S) are part of the ash or inorganic residue that remains after combustion. Carbon dioxide is produced during the burning process as a result of the oxidation of organic compounds.

(NCERT Ed. 2025-26, 11th, Page No. 109)

167. (3)

The melting point of a fatty acid is determined by two main factors:

1. Degree of Unsaturation: Unsaturated fatty acids, which contain carbon-carbon double bonds, have a lower melting point than saturated fatty acids of the same length. The double bonds create "kinks" in the hydrocarbon chain, preventing the molecules from packing tightly together. The more double bonds, the lower the melting point.
2. Chain Length: Within the same class (saturated or unsaturated), a shorter carbon chain results in a lower melting point because there are fewer points for van der Waals forces to act.

Analysing the given fatty acids:

- Linolenic acid: 18 carbons, 3 double bonds (highly unsaturated)
 - Palmitic acid: 16 carbons, saturated
 - Stearic acid: 18 carbons, saturated
1. Linolenic acid has the lowest melting point because it is polyunsaturated (3 double bonds). The kinks in its structure prevent tight packing.
 2. Palmitic acid and Stearic acid are both saturated. To rank them, we compare their chain lengths. Palmitic acid (16C) is shorter than Stearic acid (18C), so it has a lower melting point.
 3. Stearic acid has the highest melting point because it is saturated and has a longer chain length than Palmitic acid.

Therefore, the correct order from lowest to highest melting point is: Linolenic acid < Palmitic acid < Stearic acid.

(NCERT Ed. 2025-26, 11th, Page No. 106)

168. (3)

- All elements found in living tissues are also present in the earth's crust. The difference lies in their relative abundance.
- Living organisms are primarily composed of carbon, hydrogen, oxygen, and nitrogen, with a much higher proportion of carbon and hydrogen compared to the earth's crust.
- Trichloroacetic acid (TCA) is commonly used to grind living tissue. The resulting slurry is then filtered, separating the small acid-soluble molecules (micromolecules) from the large acid-insoluble molecules (macromolecules).
- The ash test involves burning the living tissue to oxidize all the organic compounds. This process destroys the organic molecules, leaving behind an inorganic residue called ash.

(NCERT Ed. 2025-26, 11th, Page No. 104)

169. (1)

The first digit of the EC number indicates the main class of the enzyme. In this case, the number is 2.7.1.1, so the enzyme belongs to Class 2.

This class is for transferases. Transferases are enzymes catalysing a transfer of a group, G (other than hydrogen) between a pair of substrate S and S'.
Glucose + ATP → Glucose-6-Phosphate + ADP: This reaction involves the transfer of a phosphate group from ATP to glucose. This is a Transferase (EC Class 2).

A dipeptide + H₂O → Two amino acids: This reaction shows a peptide bond in a dipeptide being broken by the addition of water. This is a classic example of hydrolysis, catalyzed by a Hydrolase (EC Class 3).

Pyruvate → Acetaldehyde + CO₂: This reaction involves the cleavage of a C–C bond without the use of water, resulting in the removal of a group. This is catalyzed by a Lyase (EC Class 4).

S reduced + S' oxidised → S oxidised + S' reduced: This reaction involves oxidoreduction between two substrates. This is catalysed by oxidoreductases (EC class 1)

(NCERT Ed. 2025-26, 11th, Page No. 117)

170. (1)

A quaternary structure refers to the arrangement of multiple folded polypeptide subunits. Haemoglobin is a classic example, as it is composed of four separate polypeptide chains.

Adult haemoglobin (HbA) has four subunits but only two types: α₂β₂ (two alpha and two beta chains), not four different types.

(NCERT Ed. 2025-26, 11th, Page No. 112)

171. (3)

The formula for alanine is C₃H₇NO₂.

For three molecules, the total number of atoms would be:

- Carbon: 3×3=9
- Hydrogen: 7×3=21
- Nitrogen: 1×3=3
- Oxygen: 2×3=6

Total formula (unbonded): C₉H₂₁N₃O₆

Subtract the atoms removed during peptide bond formation.

A tripeptide is formed from three amino acids. This process requires the formation of two peptide bonds. Each peptide bond is formed by a dehydration reaction, where one water molecule (H₂O) is removed.

Therefore, two water molecules (2H₂O) are removed from the total. This corresponds to the removal of 4 hydrogen atoms and 2 oxygen atoms. Subtract the removed atoms from the initial total:

- Carbon: 9–0=9
- Hydrogen: 21–4=17
- Nitrogen: 3–0=3
- Oxygen: 6–2=4

The final molecular formula of the tripeptide is C₉H₁₇N₃O₄.

(NCERT Ed. 2025-26, 11th, Page No. 109)

172. (4)

List-I	List-II
GLUT-4	Enables glucose transport in cells
Antibody	Fights infectious agents
Silicon	27.7% of weight of Earth crust
NaCl, CaCO ₃ , PO ₄ ³⁻ , SO ₄ ²⁻	Represents inorganic constituents of living tissue

(NCERT Ed. 2025-26, 11th, Page No. 109)

173. (4)

- Optimum pH is the specific pH value at which an enzyme-catalyzed reaction proceeds at the fastest rate.
- Optimum temperature is the specific temperature at which an enzyme-catalyzed reaction proceeds at the fastest rate.

(NCERT Ed. 2025-26, 11th, Page No. 116)

174. (3)

An amino acid is considered **basic** if it has an extra amino (–NH₂) group in its side chain (R group) in addition to the one attached to the alpha-carbon. This first structure is of Glycine. Its R group is a single hydrogen atom (H). It is a neutral amino acid. This second structure is of Aspartic acid. Its R group contains an extra carboxyl (–COOH) group. It is an acidic amino acid.

In third structure R group contains an additional amino (–NH₂) group. This extra amino group can accept a proton, making the amino acid basic.

This fourth structure is of Serine. Its R group is a hydroxyl methyl group (–CH₂OH). It is a neutral amino acid.

(NCERT Ed. 2025-26, 11th, Page No. 106)

175. (3)

Biomolecules are fundamentally organic compounds. Their structure is based on a backbone of carbon atoms, which form stable covalent bonds with other elements like hydrogen, oxygen, and nitrogen.

Biomolecules range from simple molecules (monomers), such as glucose and amino acids, to complex macromolecules (polymers), such as polysaccharides (starch), nucleic acids (DNA), and proteins.

Biomolecules include all molecules present in living organisms, such as carbohydrates (sugars, starch), lipids (fats, oils), nucleic acids (DNA, RNA), and proteins.

Biomolecules perform all the critical functions of life. For example, proteins act as enzymes for metabolism and provide structural support; carbohydrates are the primary energy source; lipids form cell membranes; and nucleic acids store genetic information.

(NCERT Ed. 2025-26, 11th, Page No. 105)

176. (3)

In a polysaccharide chain (say glycogen), the right end is called the reducing end and the left end is called the non-reducing end. Insulin is a protein, and its left end is the N-terminal.

Starch is the primary form of energy storage in plants. It is a polymer of glucose and is synthesized during photosynthesis to be stored in various plant parts, such as roots, seeds, and tubers.

(NCERT Ed. 2025-26, 11th, Page No. 110)

177. (2)

Chitin is a homopolymer of N-acetyl-D-glucosamine monomers. A homopolymer is a polymer made up of a single type of monomer unit. Chitin's primary function is structural, not for energy storage. Energy storage in organisms is typically handled by polysaccharides like starch (in plants) and glycogen (in animals).

Chitin provides rigidity and protection. It is the main component of the cell walls of fungi and the hard outer skeletons of arthropods, such as insects, crustaceans, and spiders.

Chitin is a polysaccharide (a carbohydrate), not a protein. Cell membranes are primarily composed of a phospholipid bilayer, not chitin.

(NCERT Ed. 2025-26, 11th, Page No. 111)

178. (2)

List-I	List-II
Niacin	Present in coenzyme NAD
Zinc	Co-factor for carboxypeptidase
Glucose	Have same number of carbons as in fructose
Cytoplasm of cell	Represented by acid soluble pool

(NCERT Ed. 2025-26, 11th, Page No. 118)

179. (3)

The general rule of thumb for chemical reactions is that for every 10°C increase in temperature, the reaction rate approximately doubles.

The temperature increases from 30°C to 60°C a total increase of 30°C.

Since the rate doubles every 10°C, a 30°C increase means the rate will double three times (30°C / 10°C = 3).

The initial rate is 2.4molL⁻¹. After the first doubling: 2.4×2=4.8

After the second doubling: 4.8×2=9.6

After the third doubling: 9.6×2=19.2

(NCERT Ed. 2025-26, 11th, Page No. 113)

180. (3)

Michaelis-Menten constant (K_m) represents the substrate concentration at which the reaction velocity is exactly half of the maximum possible velocity (V_{max}). This relationship is described by the Michaelis-Menten equation.

The K_m value does not represent the point of full saturation. Full saturation occurs at very high substrate concentrations, where the reaction rate approaches V_{max} . The K_m value, instead, is the substrate concentration required to reach half of the maximum velocity, and it is an inverse measure of the enzyme's affinity for its substrate. A low K_m indicates high affinity because a small amount of substrate is needed to reach half the maximum rate. A high K_m indicates low affinity.

(NCERT Ed. 2025-26, 11th, Page No. 116)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>